

Practical -09

1) File I/O

```
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GNU nano 7.2 fileio.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;
    char buffer[100];

    // Low-level I/O
    fd = open("data.txt", O_RDONLY);

    int n = read(fd, buffer, sizeof(buffer) - 1);
    buffer[n] = '\0';

    printf("Using open/read:\n%s\n", buffer);

    close(fd);

    // Standard I/O
    FILE *fp = fopen("data.txt", "r");

    printf("Using fopen/fread:\n");
    n = fread(buffer, 1, sizeof(buffer) - 1, fp);
    buffer[n] = '\0';

    printf("%s\n", buffer);

    fclose(fp);

    return 0;
}
```

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bharath_sai_reddy@LAPTOP-1  ×  +  ▾
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ nano fileio.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ echo "Hello OSSP File I/O" > data.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ gcc fileio.c -o fileio
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ ./fileio
Using open/read:
Hello OSSP File I/O

Using fopen/fread:
Hello OSSP File I/O
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$
```

2) redirect

```
bharath_sai_reddy@LAPTOP-1  ×  +  ▾  
GNU nano 7.2                                                                redirect.c  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main() {  
    int fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0666);  
  
    dup2(fd, STDOUT_FILENO);  
  
    printf("This output goes into output.txt\n");  
  
    close(fd);  
  
    return 0;  
}
```

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bharath_sai_reddy@LAPTOP-1  ×  +  ▾  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ nano redirect.c  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ gcc redirect.c -o redirect  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ ./redirect  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-09$ cat output.txt  
This output goes into output.txt
```